CCPBioSim

VMD for Visualisation

A practical guide for beginners

Your Starting Structure: PDB Search

Go to the protein data bank (type pdb into google).

Before you start, scroll down and have a look at "Molecule of the month". If you click on "Structural view of biology" you can access the archive. This is highly recommended reading!

In the pdb search facility, find human ubiquitin 1UBQ.pdb (X-ray crystal structure)

There is a "pdb redo" version of 1UBQ.pdb – so we should use that structure instead (called 1ubq_final.pdb).

And 1D3Z.pdb (NMR) by typing in the codes.

Download the files as "pdb format".

VMD for Visualisation ~ X-ray Structure

1. Start VMD
In Linux you can type VMD at the command line, in windows you will need to select VMD from the start menu.

2. Load your X-ray pdb file into VMD

File/New Molecule/Browse/
Pick 1ubq_final.pdb
OK and Load.

VMD ~ “Hot” Keys

VMD has particular “hot keys” that make the program easy to use.

“r” Rotate structure
“t” Translate structure
“s” Scale structure (eg zoom in or out)
“1” Label an atom
“2” Measure a bond between 2 atoms
“3” Measure the angle between 3 atoms
“4” Measure the dihedral between 4 atoms

3. Play with the different representations (e.g. *VdW*, *new cartoon*). Display the backbone only. *Delete "all" and type "backbone" in the selected atoms box*

When you colour the structure by "*name*" (default), notice which elements have which colours (e.g. hydrogens are white).

What else is there in the file, apart from the protein?

4. In *new cartoon* representation, colour the protein by "*secondary structure*".

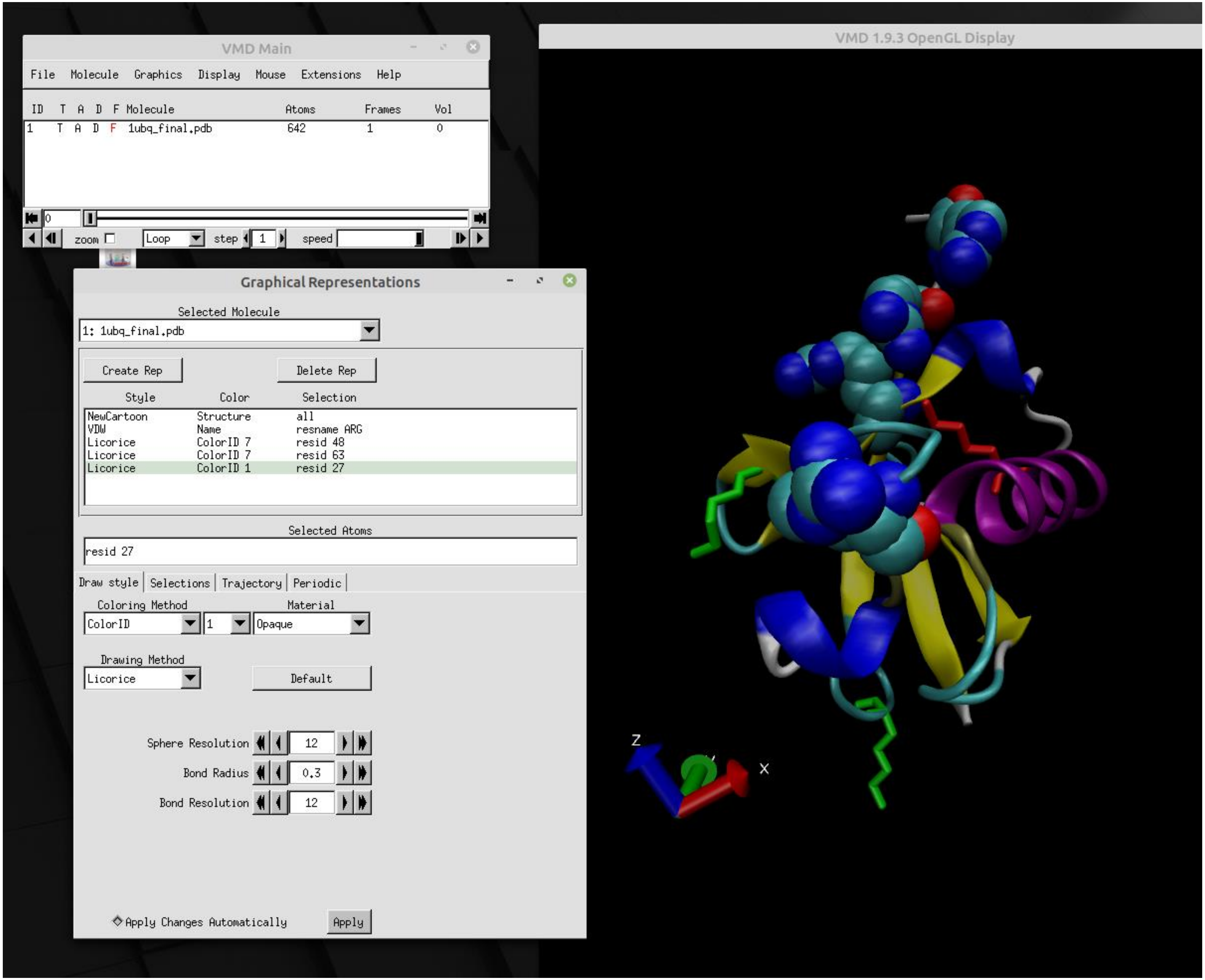
5. Make the ARG residues VdW and colour these by "*name*".

6. Make LYS 63 and 48 green but LYS 27 red.

7. Put the molecule in *cartoon* representation. Display all of the *charged* residues using *VdW*.

Selection/coloring method/beta

What do you notice about the location of the charged residues?


VMD Main
File Molecule Graphics Display Mouse Extensions Help
ID T A D F Molecule Atoms Frames Vol
1 T A D F 1ubq_final.pdb 642 1 0
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Create Rep
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Licorice ColorID 1 resid 27
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VMD 1.9.3 OpenGL Display
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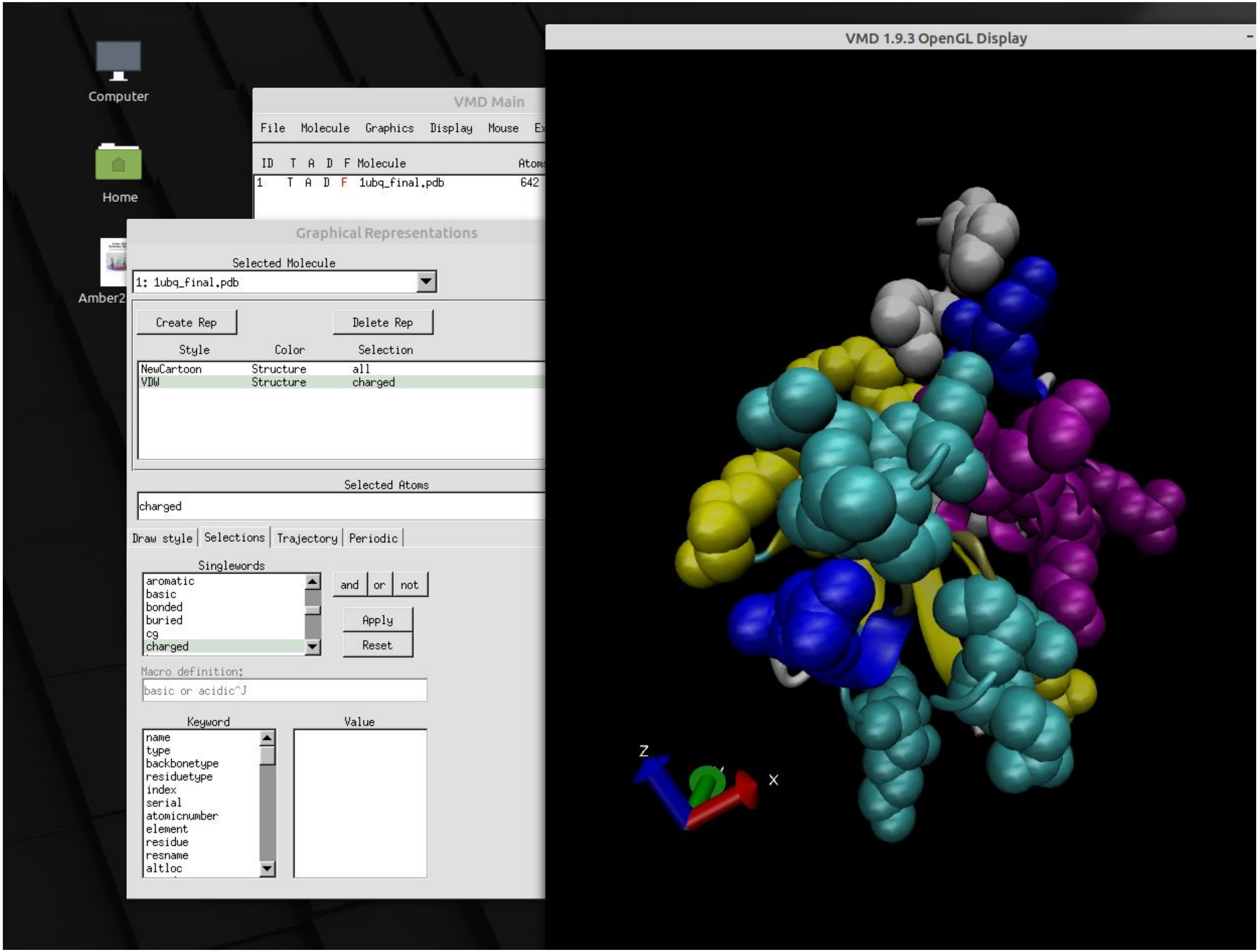

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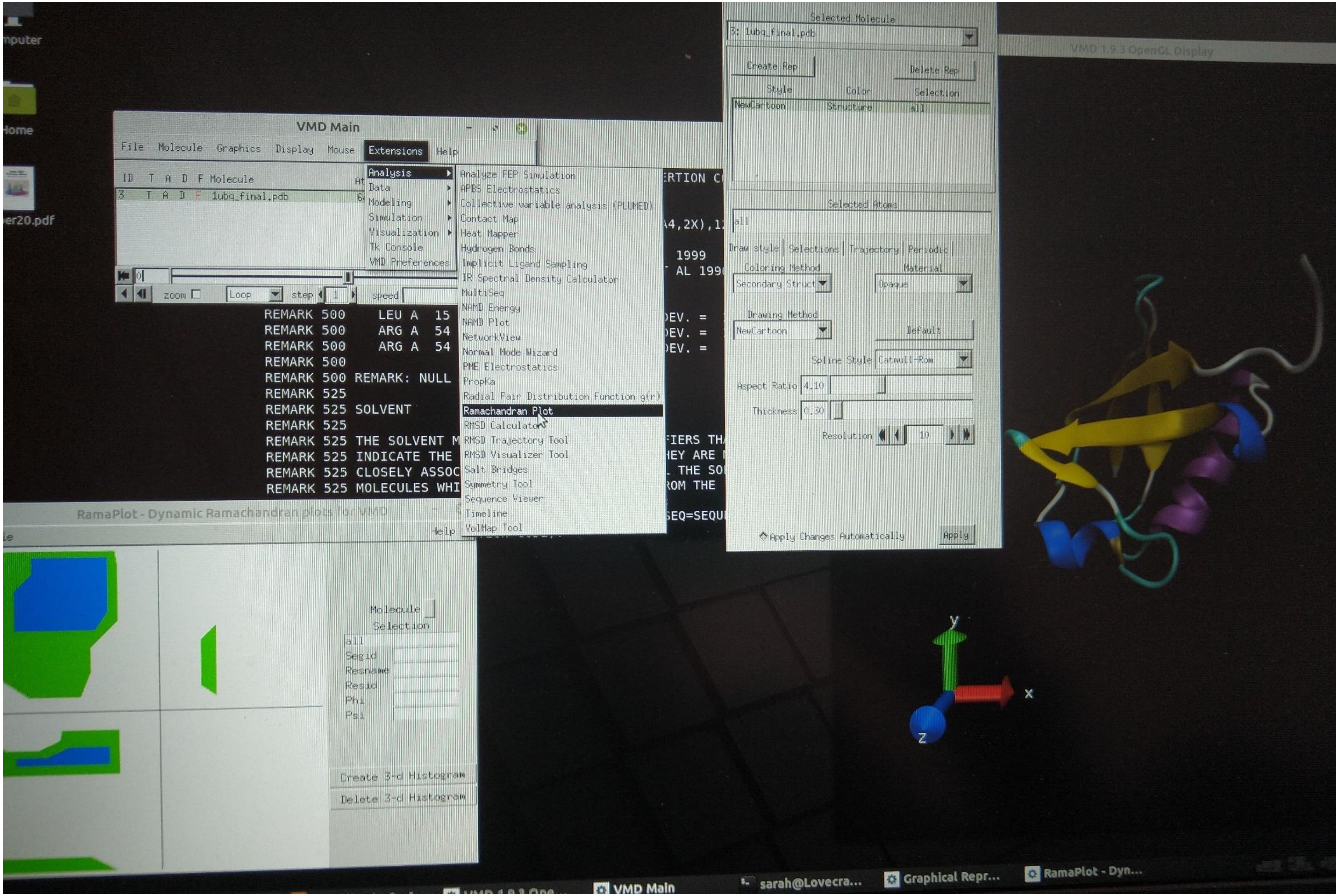
8. Colour the protein by B-factor (the largest B-factors show the most flexible regions).

Draw style/coloring method/beta

9. Plot a Ramachandran plot

Extensions/analysis

In the "*VMD Main*" window, you will need to click on "*1ubq_final.pdb*" to select your structure.



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Visualization
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VMD Preferences
Analyze FEP Simulation
APBS Electrostatics
Collective variable analysis (PLUMED)
Contact Map
Heat Mapper
Hydrogen Bonds
Implicit Ligand Sampling
IR Spectral Density Calculator
MultiSeq
NAMD Energy
NAMD Plot
NetworkView
Normal Mode Wizard
PME Electrostatics
PropKa
Radial Pair Distribution Function g(r)
Ramachandran Plot
RMSD Calculator
RMSD Trajectory Tool
RMSD Visualizer Tool
Salt Bridges
Symmetry Tool
Sequence Viewer
Timeline
VolMap Tool
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Visualising NMR Structures with VMD

1. Delete the X-ray structure and read the NMR structure into VMD.

2. What two very obvious differences are there with the X-ray structure?

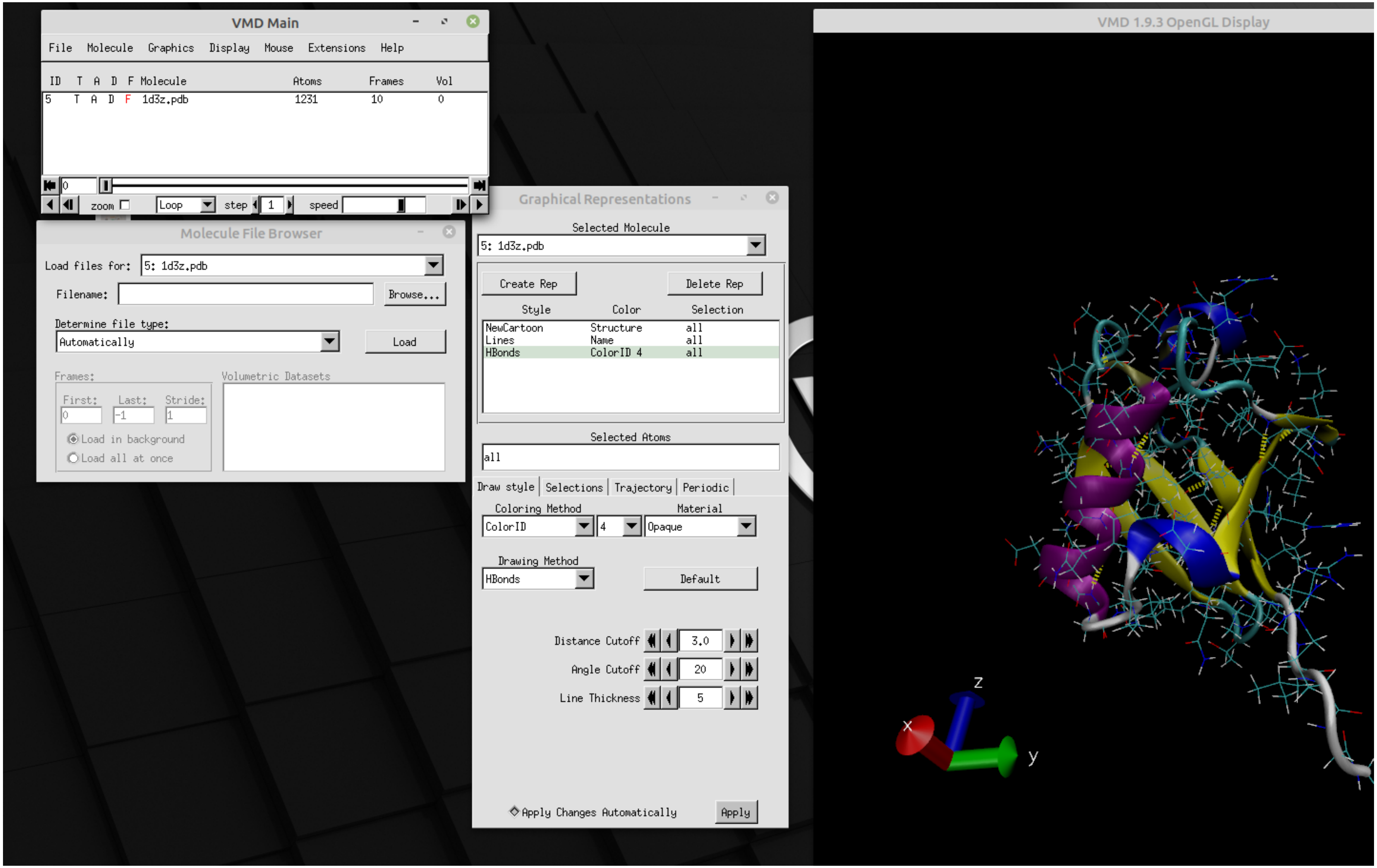
3. Compare the regions of highest flexibility with those you detected using crystallographic B-factors.

Visualising hydrogen bonds

4. Create a new representation and make this *new cartoon*. Colour by *secondary structure*.

5. Display the hydrogen bonds. Colour these "yellow" and adjust the line thickness to 5.

Where do most of the hydrogen bonding interactions occur, and do they persist for all conformers?


VMD Main
File Molecule Graphics Display Mouse Extensions Help
ID T A D F Molecule Atoms Frames Vol
5 T A D F 1d3z.pdb 1231 10 0
zoom Loop step 1 speed
Molecule File Browser
Load files for: 5: 1d3z.pdb
Filename:
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Determine file type:
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Selected Molecule
5: 1d3z.pdb
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all
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Material
ColorID 4 Opaque
Drawing Method
HBonds
Default
Distance Cutoff 3.0
Angle Cutoff 20
Line Thickness 5
Apply Changes Automatically
Apply
VMD 1.9.3 OpenGL Display
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Plotting Inter-atomic Distances

6. Display the N-terminal (first) and C-terminal (last) residues as *liquorice*.

7. Display the bond distance between the Cα of the N-terminal (first) and C-terminal (last) residues.

Press “2” (for bonds)

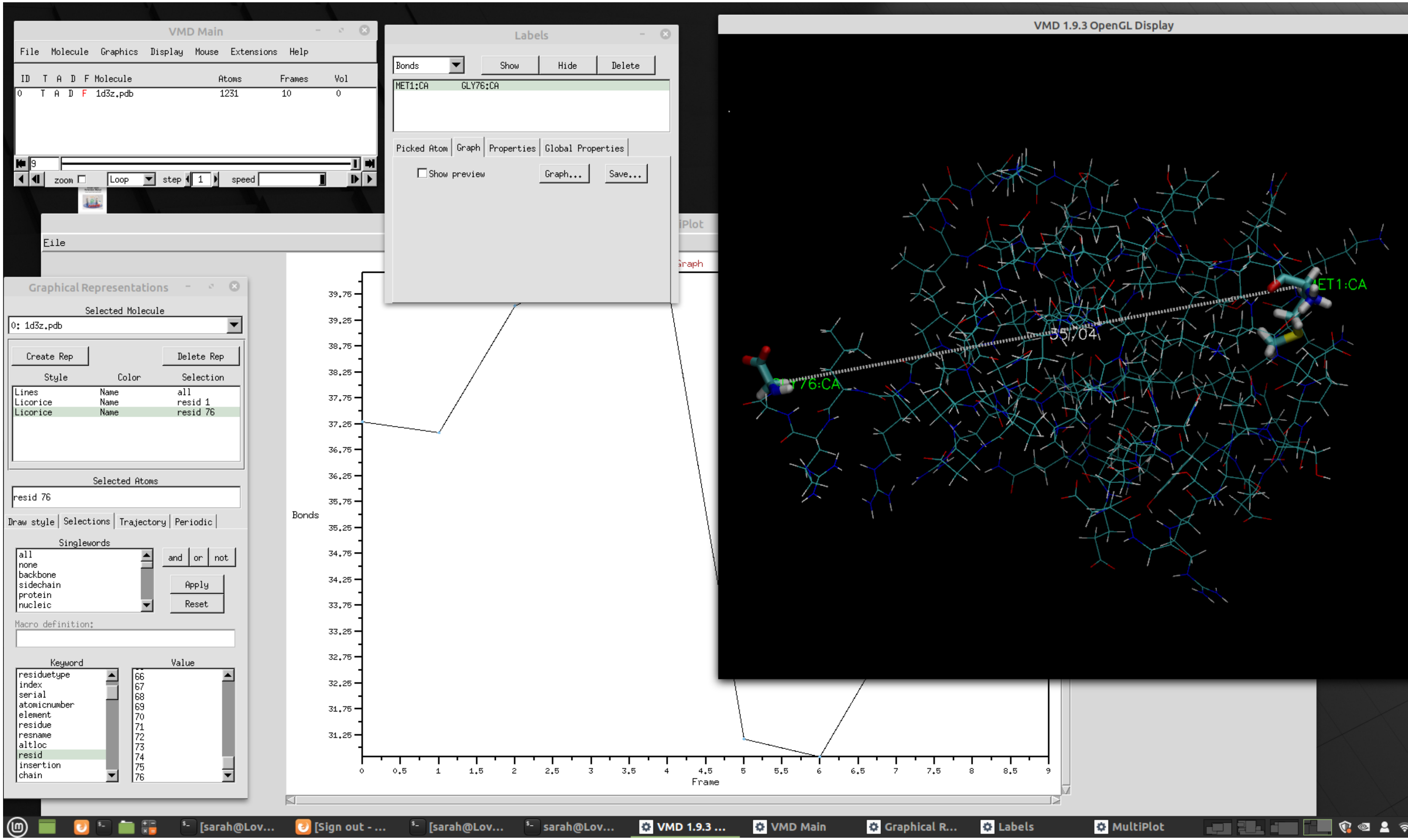
Click on one Cα atom then the other

8. How does this distance change throughout the ensemble? Plot a graph of your results.

Graphics/labels/bonds

Click on the bond

Select graph



VMD Main
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VMD 1.9.3 OpenGL Display
MET1:CA
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Selected Molecule
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Create Rep Delete Rep
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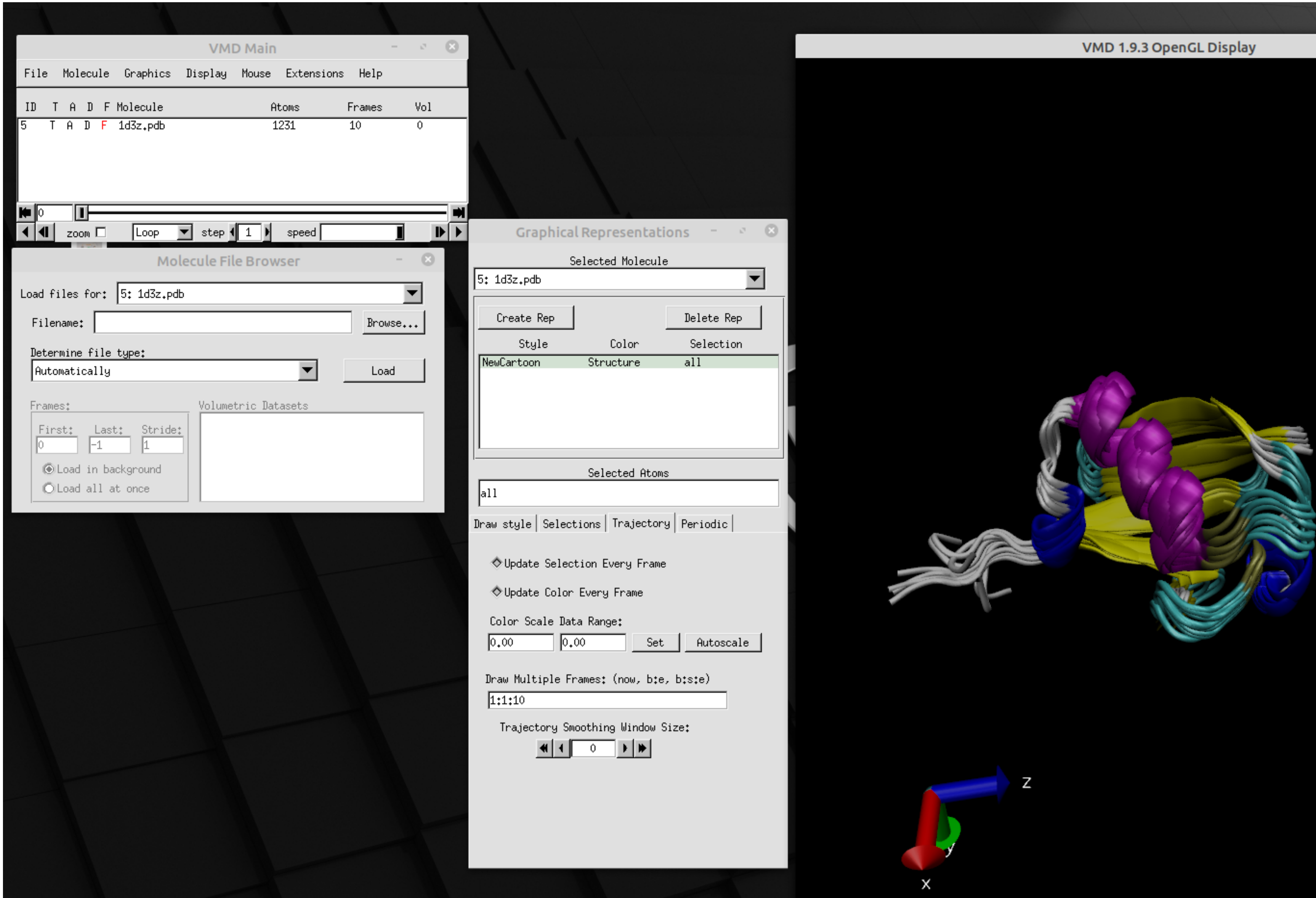
[sarah@Lov... [Sign out - ... [sarah@Lov... sarah@Lov... VMD 1.9.3 ... VMD Main Graphical R... Labels MultiPlot

Visualising Conformational Diversity

9. Delete all of the bond and atom labels.

10. Visualise the conformational diversity of the NMR ensemble by displaying all of the structures simultaneously.

Trajectory/Draw Multiple Frames


VMD Main
File Molecule Graphics Display Mouse Extensions Help
ID T A D F Molecule Atoms Frames Vol
5 T A D F 1d3z.pdb 1231 10 0
zoom Loop step 1 speed
Molecule File Browser
Load files for: 5: 1d3z.pdb
Filename: Browse...
Determine file type:
Automatically Load
Frames: Volumetric Datasets
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0 -1 1
Load in background
Load all at once
Graphical Representations
Selected Molecule
5: 1d3z.pdb
Create Rep Delete Rep
Style Color Selection
NewCartoon Structure all
Selected Atoms
all
Draw style Selections Trajectory Periodic
Update Selection Every Frame
Update Color Every Frame
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1:1:10
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VMD 1.9.3 OpenGL Display
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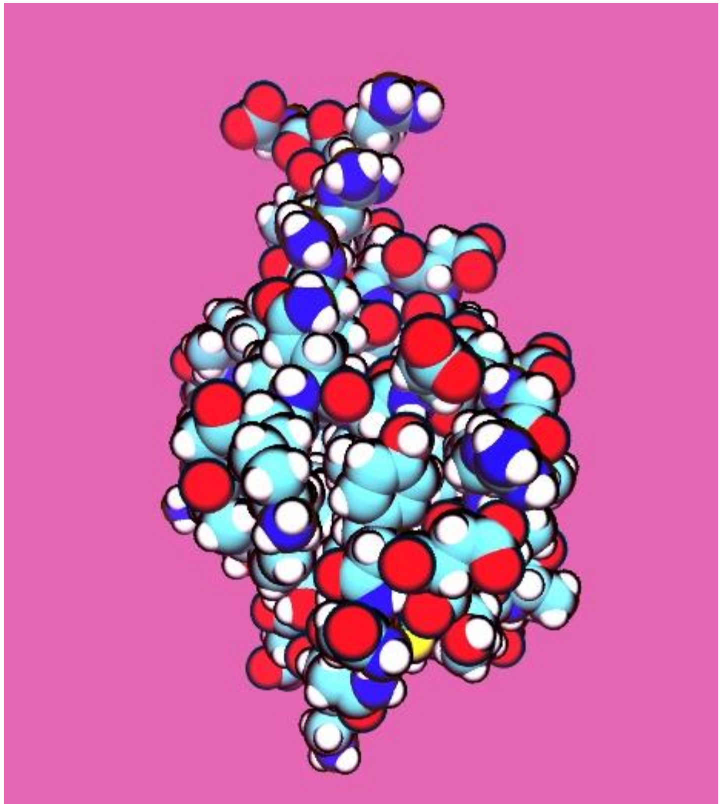
Make a Pretty Picture

1. Go to *Display/display settings* and turn on *Shadows* and *Amb occl*.
2. Under *Graphics/colors/categories/display* choose *background*, and pick a lurid colour (e.g. lime green).
3. Using *Graphics/representations/trajectory* draw multiple frames. Represent your protein using *tubes*.
4. Colour your protein by *timestep*.
5. Switch the axes display off.
6. Do *File/render*. Choose the *tachyon internal* render option.
7. Rename your file “vmdscene.tga” to something less generic.

You will need to use a conversion program (like gimp/imagemagic) to make these compatible with powerpoint.

8. Now display only one structure using the VdW representation.
9. Colour by atom name.
10. Under *material* select *Goodsell*.
11. Render your image!!



Cheat Sheet!!

X-ray structure:

1. You will be able to see lots of red dots surrounding the protein. These are crystalline waters – but of course the hydrogens are missing as they are not resolved.
2. You will notice that the charged residues are mainly located on the outside of the protein, where they are exposed to solvent. If you display the hydrophobic residues (under *selections/singlewords/hydrophobic*) you will see that these are predominantly buried within the protein core.

NMR structure:

1. You will see that hydrogens are present in the NMR structure, and that it contains multiple conformers. You will also notice that there are no crystalline waters.
2. The hydrogen bonds are mainly formed between backbone residues in the secondary structure elements, e.g. α-helices and β-sheets. These do not persist throughout each conformer – rather – they "flicker" on and off as the protein explores its conformational ensemble.